

klasično: $x(0) \rightarrow$ napovemo $x(t) \forall t > 0$

kvantno: imamo $\Psi(x, 0) \rightarrow$ lahko napovemo le verjetnost, da je delec ob času $t \dots$

(verjetnosti za naključno detekcijo delca in ustreznih opazljivk)

$\rightarrow$ potrebujemo PDE (parc. dif. en.), ker $\Psi = \Psi(x, t)$

Prost delec (v prostoru $\approx V=0$)
potencial

- pokazujemo z val. paketom $\Psi(x, t) = \int_{-\infty}^{\infty} A(k) e^{i(kx - \omega t)} dk$

disperzijska relacija: $E = \hbar \omega = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$

$E \uparrow \propto k^2 \uparrow$
 ima zvezo s $\frac{\partial}{\partial t}$ ima zvezo s $\frac{\partial^2}{\partial x^2}$

$$\Psi(x, t) = \int_{-\infty}^{\infty} A(k) \exp(i[kx - \frac{\hbar k^2}{2m} t]) dk$$

$$\frac{\partial^2 \Psi}{\partial x^2} = \int_{-\infty}^{\infty} A(k) (ik)^2 \exp(\dots) dk$$

$$\frac{\partial \Psi}{\partial t} = \int_{-\infty}^{\infty} A(k) \left(-\frac{i\hbar}{2m}\right) k^2 \exp(\dots) dk$$

$$\frac{-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2}}{\frac{p^2}{2m}} = i\hbar \frac{\partial \Psi}{\partial t} \quad \quad \quad \frac{p^2}{2m} = E$$

$p \rightarrow \left(-i\hbar \frac{\partial}{\partial x}\right)^2$ $E \xrightarrow{\text{"predpis"}} i\hbar \frac{\partial}{\partial t} \xrightarrow{\text{"deluje"}} \Psi(x, t)$
operator

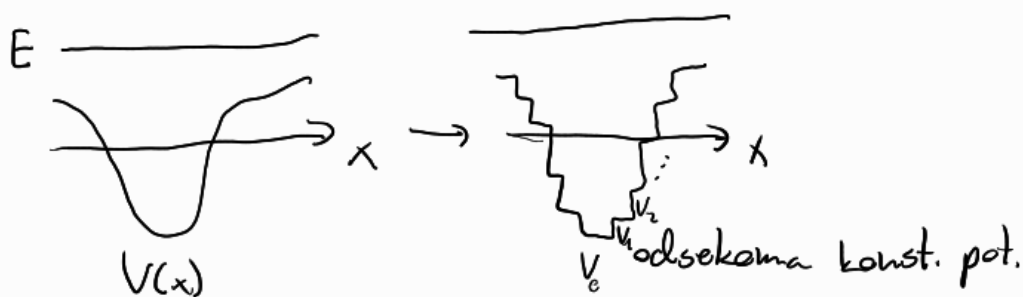
Prostor $\approx V \neq 0$: $E = \underbrace{\frac{p^2}{2m}}_{\text{kin.}} + \underbrace{V}_{\text{pot.}}$
 $V = \text{konst.}$

za ilustracijo: $\Psi(x, t) = Ae^{i(kx - \omega t)}$

disperz. relacija: $E = \hbar\omega = \frac{\hbar^2 k^2}{2m} + V$

$$\rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + \underbrace{V\Psi}_{\text{multiplikativni operator}} = i\hbar \frac{\partial \Psi}{\partial t}$$

$V(x)$ poljubni:



$$E = \hbar\omega = \frac{\hbar^2 k_0^2}{2m} + V_0 = \frac{\hbar^2 k_1^2}{2m} + V_1 = \dots$$

$\rightarrow$ $-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, t)}{\partial x^2} + V(x)\Psi(x, t) = i\hbar \frac{\partial \Psi}{\partial t}$ Nestacionarna (čas. odv.)
 Schrödingerjeva enačba (NSE)

oziroma

$$\hat{H}\Psi = i\hbar \frac{\partial \Psi}{\partial t}$$

$$\hat{H} = \hat{T} + \hat{V}$$

$\uparrow$
Hamiltonov
operator
celotne E

$\downarrow$ $\wedge$ pomeni
operator

Oponaba: $\sin(kx - \omega t), \cos(kx - \omega t), e^{\pm i(kx - \omega t)}$

so rešitve običajne val. en., ker imamo $\frac{\partial^2 u}{\partial t^2}$ in $\frac{\partial^2 u}{\partial x^2}$

→ NSE: ker $i\hbar \frac{\partial \Psi}{\partial t} \in \mathbb{R} \Rightarrow \Psi \in \mathbb{C}$

upr. $\Psi(x, t) = \frac{A}{2} [e^{i(kx - \omega t)} + e^{i(-kx - \omega t)}] = A \cos kx e^{-i\omega t}$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi = \left(\frac{\hbar^2 k^2}{2m} + V \right) A \cos kx e^{-i\omega t}$$

$$i\hbar \frac{\partial \Psi}{\partial t} = i\hbar (-i\omega) A \cos kx e^{-i\omega t}$$

je enako, če velja: $\hbar\omega = \frac{\hbar^2 k^2}{2m} + V$ ✓

Verjetnostna interpretacija Ψ

(kako iz $\Psi(x, t)$ dobiti verjetnost za detekcijo)

- Ψ ni opazljiva, ker je $\mathbb{C}$

$\rho(x, t) = |\Psi|^2 = \Psi \Psi^*$ verjetnostna gostota, ki ga opisuje VF Ψ

$$P(x_1 \leq x_{\text{detekc}} \leq x_2) = \int_{x_1}^{x_2} \rho(x, t) dx \quad \text{ob času } t$$

Normalizacija: deleč gotovo najdemo na $(-\infty, \infty)$

$$\int_{-\infty}^{\infty} \rho(x, t) dx = 1 \leftarrow \text{samo dogovor, da se izognemo dvoumnosti glede faktorja}$$

oz. po ustroznem D_f

(Ψ reši NSE $\Rightarrow a\Psi$ reši NSE)

ρ je v splošnem odv. od t :

NSG v prostoru $V \approx 0$:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = i\hbar \frac{\partial \Psi}{\partial t} \xrightarrow{\text{konjugirano}} -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi^*}{\partial x^2} = -i\hbar \frac{\partial \Psi^*}{\partial t}$$

$$\frac{\partial}{\partial t} (\underbrace{\Psi \Psi^*}_{\rho}) = \Psi^* \frac{\partial \Psi}{\partial t} + \frac{\partial \Psi^*}{\partial t} \Psi = \Psi^* \left(-\frac{\hbar}{2im} \frac{\partial^2 \Psi}{\partial x^2} \right) + \left(\frac{\hbar}{2im} \frac{\partial^2 \Psi^*}{\partial x^2} \right) \Psi =$$

$$= -\frac{\hbar}{2im} \left(\Psi^* \frac{\partial^2 \Psi}{\partial x^2} - \frac{\partial^2 \Psi^*}{\partial x^2} \Psi \right) = -\frac{\partial}{\partial x} \left[\frac{\hbar}{2im} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) \right]$$

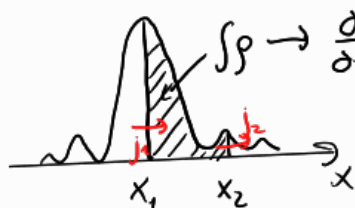
$j(x,t)$ gost. verj. toka

$$\boxed{\frac{\partial \rho}{\partial t} + \frac{\partial j}{\partial x} = 0}$$

kako se spr.
verj. gost.

koliko verj.
"steče" iz
krajnjega
intervala (oz.
"priteče" vanj)

$$\frac{\partial}{\partial t} \int_{x_1}^{x_2} \rho(x,t) dx = \int_{x_1}^{x_2} \frac{\partial \rho}{\partial t} dx = - \int_{x_1}^{x_2} \frac{\partial j}{\partial x} dx = j_1(x,t) - j_2(x,t)$$



$\int \rho \rightarrow \frac{\partial \rho}{\partial t}$ podano z razliko
tokov pri x_1 in x_2

Vse velja tudi za $V \neq 0$, če $V \in \mathbb{R}$ ($V = V^*$)

Subtilna poanta:

$$\int_{-\infty}^{\infty} \rho(x,t) dx = 1 \quad \uparrow \quad \text{neodv. od } t?$$

$$\frac{\partial}{\partial t} \int_{x_1}^{x_2} \Psi^* \Psi dx = -\frac{\hbar}{2im} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) \Big|_{x_1}^{x_2} = 0$$

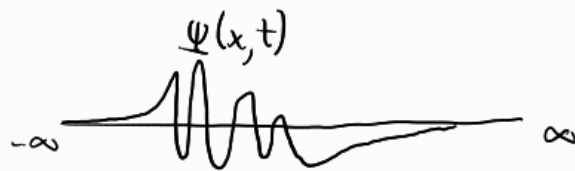
↑ želimo, če

$$\int_{-\infty}^{\infty} \rho dx = \frac{\partial}{\partial t} 1 = 0$$

↑ kaj bo

→ ko $x_1 \rightarrow -\infty$ in $x_2 \rightarrow \infty$, mora iti $\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \rightarrow 0$

✓ običajno izpolnjeno



→ Sprejemljive VF:

- Ψ zvezne (ker ne moremo imeti skokov v verjetnosti za detekcijo)
- $\frac{\partial \Psi}{\partial x}$ zvezen (ker Schrödingerjeva en. 2. reda), razen ko $V \rightarrow \infty$

Stacionarna stanja

(separacija krajevne in časovne koord.)

Nastavek za rešitev NSE:
(ansatz)

$$\Psi(x, t) = \psi(x) f(t)$$

$$\rightarrow -\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} f + V \psi f = i\hbar \psi \frac{df}{dt} \quad / : \psi f$$

$$\frac{-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V \psi}{\psi} = \frac{i\hbar \frac{df}{dt}}{f}$$

- če $V = V(x)$, je leva str. odvisna samo od x , desna pa od t

$$\Rightarrow \frac{-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V \psi}{\psi} = \frac{i\hbar \frac{df}{dt}}{f} \equiv \Lambda = \text{konst.}$$

↑
lambda

(separacijska konst.)

→ dobimo dve navadni dif. enačbi:

$$\textcircled{1} \quad i\hbar \frac{df}{dt} = \Lambda f$$

$$\textcircled{2} \quad -\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V \psi = \Lambda \psi$$

Rešitev ①

$$f(t) = e^{-i\Lambda t/\hbar} = \cos\left(\frac{\Lambda t}{\hbar}\right) - i \sin\left(\frac{\Lambda t}{\hbar}\right) \rightarrow \frac{\Lambda}{\hbar} = \omega \rightarrow \Lambda = \hbar\omega = E$$

↑
! separacijska konst. je E delca

→ ②

$$-\frac{\hbar^2}{2m} \frac{d^2 \Psi}{dx^2} + V(x)\Psi = E\Psi$$

STAC SCHRÖDINGERJEVA
ENAČBA (SSE)

oz. $\hat{H}\Psi = E\Psi$

- za nekaj $V(x) \exists$ točne rešitve:

- $V=0$ pot. jama
- $V \propto x^2$ harm. oscilator ~~klasičen~~ (v kvantnem svetu)
- $V \propto x^3$
- ⋮

⇒ rešitev: $\Psi(x,t) = \overset{\text{težko}}{\downarrow} \Psi(x) \overset{\text{lahko}}{f(t)}$

$\hat{H}\Psi = E\Psi$ je problem lastnih vrednosti
/ lastna funkcija (energije)
lastna vrednost (energije)

spekter

- reševanje SSE privede do nabora lastnih vr. $\{E_1, E_2, \dots\}$ in lastnih funke. $\{\Psi_1, \Psi_2, \dots\}$

- tudi $\Psi(x)$ so normirani: $\int_{-\infty}^{\infty} |\Psi(x)|^2 dx = 1$ (ker f eksponentna)

stanja ($\Psi(x,t) = \Psi(x)f(t)$), za katera velja $|\Psi(x,t)|^2 = |\Psi(x)|^2$, so STACIONARNA.

→ za $\Psi(x,t) = \Psi(x)e^{-iE_1 t/\hbar}$ je to avtomatsko izpolnjeno.

Zgled: $\Psi_1(x, t) = \Psi_1(x) e^{-i\epsilon_1 t/\hbar}$, $\Psi_2(x, t) = \Psi_2(x) e^{-i\epsilon_2 t/\hbar}$, ...

Ψ_1, Ψ_2 sta stac. stanj:

Recimo, da je dejansko stanje deka $\Psi(x, t) = a_1 \Psi_1(x, t) + a_2 \Psi_2(x, t)$

$$|\Psi(x, t)|^2 = |a_1|^2 \underbrace{|\Psi_1|^2}_{=|\Psi_1|^2} + |a_2|^2 \underbrace{|\Psi_2|^2}_{=|\Psi_2|^2} + a_1^* a_2 \underbrace{\Psi_1^* \Psi_2}_{=e^{-i(\epsilon_2 - \epsilon_1)t/\hbar}} + a_1 a_2^* \underbrace{\Psi_1 \Psi_2^*}_{=e^{i(\epsilon_2 - \epsilon_1)t/\hbar}}$$

-derimo, da $a_1, a_2 \in \mathbb{R}$

$$|\Psi(x, t)|^2 = \boxed{\text{neodv. od } t} + 2a_1 a_2 \cos\left(\frac{(\epsilon_2 - \epsilon_1)t}{\hbar}\right)$$

$\Rightarrow \Psi$ ni stacionarno, čeprav je lin. komb. dveh stac. stanj

$\Rightarrow$ statistične lastnosti oscilirajo s $\omega = (\epsilon_2 - \epsilon_1)/\hbar$

zgleđ: ∞ jama

Lastnosti lastnih VF

-ker VF ortogonalne: lahko jih razvijemo po lastnih funkcijah:

$$\Psi(x, 0) = \sum_n c_n \Psi_n(x) \quad (\text{podatek naloge})$$

$$\rightarrow \Psi(x, t) = \sum_n \underbrace{c_n}_{\text{red}} \underbrace{\Psi_n(x)}_{\text{red}} e^{-i \underbrace{\epsilon_n}_{\text{red}} t/\hbar}$$

Kako dobimo c_n ?

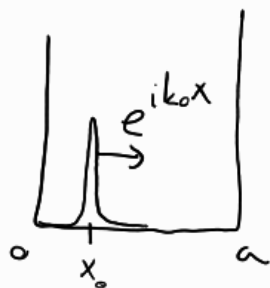
$$\int \Psi_m^*(x) \Psi(x, 0) dx = \int \Psi_m^*(x) \sum_n c_n \Psi_n(x) dx = \sum_n c_n \int \Psi_m^* \Psi_n dx = \sum_n c_n \delta_{m,n} = c_m$$

$$\rightarrow \boxed{c_n = \int \Psi_n^*(x) \Psi(x, 0) dx}$$

Zgled: valovni paket v ∞ jami

Gaussov paket

$$\Psi(x,0) = \frac{1}{(2\pi\delta^2)^{1/4}} e^{-\frac{(x-x_0)^2}{4\delta^2}} e^{ik_0 x}$$



$\Psi_n(x)$ = lastne VF za ∞ jamo

$$E_n = n^2 E_1$$

$$c_n = \int_0^a \Psi_n(x) \Psi(x,0) dx = \dots \approx i(2\pi)^{1/4} \sqrt{\frac{\delta}{a}} \left(\dots \right)$$

$\downarrow$ $\sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$ $\nearrow$ $\delta \ll a$

$\Rightarrow |\Psi(x,t)|^2$ = animacija

- paket se razširi, manjša se amplituda (disperzija)
- na robovih interferirajo vpadne in odbite komp. paketa

Komentar:

za prost delec so lastna stanja $\in$ tudi lastna stanja GK (p oz k)

$$\left. \begin{array}{l} \frac{\partial}{\partial t} \rightarrow \omega \quad (E = \hbar\omega) \\ \frac{\partial^2}{\partial x^2} \rightarrow k^2 \quad (p = \hbar k) \end{array} \right\} \begin{array}{l} \text{v } \infty \text{ jami to ne velja (delec ni prost)} \\ \text{- ne more iti skozi steno, en sam } \vec{p} \text{ ni ok} \end{array}$$

$$\Psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) = \sqrt{\frac{2}{a}} \frac{1}{2i} \left(e^{ik_n x} - e^{-ik_n x} \right) = \text{ravni val (v levo + v desno)}$$

$\rightarrow$ vsak pol časa

operator, ki deluje na Ψ na desni

$$\langle A \rangle = \int_{-\infty}^{\infty} \Psi^* A(x) \Psi dx$$

če $A(x)$ multiplikativen, je $\langle A \rangle = \int_{-\infty}^{\infty} |\Psi|^2 A(x) dx$

Pričakovana vrednost GK:

$$\langle p \rangle = m \frac{d\langle x \rangle}{dt} = m \frac{d}{dt} \int_{-\infty}^{\infty} \Psi^* x \Psi dx = m \int_{-\infty}^{\infty} \frac{\partial}{\partial t} [\Psi^* x \Psi] dx =$$

$$= m \int_{-\infty}^{\infty} \left[\frac{\partial \Psi^*}{\partial t} x \Psi + \Psi^* x \frac{\partial \Psi}{\partial t} \right] dx$$

Predp. $V \in \mathbb{R}$, pogledamo NSE:

$$\frac{\partial \Psi}{\partial t} = \frac{i\hbar}{2m} \frac{\partial^2 \Psi}{\partial x^2} + \frac{V}{i\hbar} \Psi \quad \text{in} \quad \frac{\partial \Psi^*}{\partial t} = -\frac{i\hbar}{2m} \frac{\partial^2 \Psi^*}{\partial x^2} - \frac{V}{i\hbar} \Psi^*$$

$$\rightarrow \langle p \rangle = \frac{i\hbar}{2} \int_{-\infty}^{\infty} \left(\Psi^* x \frac{\partial^2 \Psi}{\partial x^2} - \frac{\partial^2 \Psi^*}{\partial x^2} x \Psi + \cancel{\text{delava}} V \right) dx =$$

$$= \frac{i\hbar}{2} \int_{-\infty}^{\infty} x \frac{\partial}{\partial x} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \frac{\partial \Psi^*}{\partial x} \Psi \right) dx = \frac{i\hbar}{2} \left(x \Psi^* \frac{\partial \Psi}{\partial x} - x \frac{\partial \Psi^*}{\partial x} \Psi \right) \Big|_{-\infty}^{\infty} - \frac{i\hbar}{2} \int_{-\infty}^{\infty} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \frac{\partial \Psi^*}{\partial x} \Psi \right) dx$$

$$= -\frac{i\hbar}{2} \int_{-\infty}^{\infty} \Psi^* \frac{\partial \Psi}{\partial x} dx = \boxed{\int_{-\infty}^{\infty} \Psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi dx = \langle p \rangle}$$

operator GK

0, ker val. funkc. dovolj hitro padajo $\rightarrow 0$

naredimo per partes, da $\Psi^* \Psi \rightarrow \Psi^* \Psi'$

$$\boxed{\hat{p} = -i\hbar \frac{\partial}{\partial x}}$$

deluje samo na krajevno koord.

OPERATOR GK

Operator T:

$$T = \frac{p^2}{2m} \rightarrow \hat{T} = \frac{\hat{p}^2}{2m} = \frac{1}{2m} \left(-i\hbar \frac{\partial}{\partial x} \right)^2 = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \quad \text{že vemo}$$

$$\rightarrow \sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}, \quad \sigma_x = \frac{\hbar}{2\sigma_p}$$

$$\rightarrow \langle E \rangle = \langle T \rangle + \langle V \rangle = \left\langle \frac{p^2}{2m} \right\rangle + \langle V \rangle = \int_{-\infty}^{\infty} \Psi^* \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \Psi dx = \int_{-\infty}^{\infty} \Psi^* \hat{H} \Psi dx =$$

$$\langle E \rangle = \int_{-\infty}^{\infty} \Psi^* \underbrace{\left(i\hbar \frac{\partial}{\partial t} \right)}_{\hat{H}(\hat{E}) = i\hbar \frac{\partial}{\partial t}} \Psi dx$$

operator celotne E

Komentar: $E \in \mathbb{R}$

def. $|\Psi(x, t)|^2 = |\Psi(x)|^2$

$$\Psi_n(x, t) = \Psi_n(x) e^{-iE_n t/\hbar}$$

$$i\hbar \frac{\partial \Psi_n}{\partial t} = \Psi_n(x) E_n e^{-iE_n t/\hbar} = E_n \Psi_n(x, t)$$

$$\langle E \rangle = \int_{-\infty}^{\infty} \Psi_n^* \left(i\hbar \frac{\partial}{\partial t} \right) \Psi_n(x, t) dx = E_n \underbrace{\int_{-\infty}^{\infty} \Psi_n^* \Psi_n dx}_{=1 \text{ (norma)}} = E_n$$

$$\left(i\hbar \frac{\partial}{\partial t} \right)^2 \Psi_n = E_n^2 \Psi_n$$

$$\langle E^2 \rangle = \int_{-\infty}^{\infty} \Psi_n^* \left(i\hbar \frac{\partial}{\partial t} \right)^2 \Psi_n dx = E_n^2$$

$$\Delta E = \sqrt{\langle E^2 \rangle - \langle E \rangle^2} = 0 \rightarrow \text{Energija ostro določena} \xrightarrow{\text{Heisenberg}} \text{časovna nedol.} = \infty$$

$\rightarrow$ stat. stanj ne more biti $\rightarrow$ motnje ($\rightarrow$ sevaanje)

Meritev E

dejmo, da je delec v stanju $\Psi(x)$, ki ni lastno.

$$\Psi(x) = \sum_n c_n \Psi_n(x)$$

$$\begin{aligned} \Rightarrow \langle E \rangle &= \int_{-\infty}^{\infty} \Psi^* \hat{H} \Psi dx = \sum_{m,n} c_m^* c_n \int_{-\infty}^{\infty} \Psi_m^* \hat{H} \Psi_n dx = \sum_{m,n} c_m^* c_n E_n \underbrace{\int_{-\infty}^{\infty} \Psi_m^* \Psi_n dx}_{\delta_{m,n}} = \\ &= \sum_{m,n} c_m^* c_n E_n \delta_{m,n} = \sum_n |c_n|^2 E_n \end{aligned}$$

Stanje Ψ mora biti normirano: $1 = \int_{-\infty}^{\infty} \Psi^* \Psi dx = \sum_{m,n} c_m^* c_n \delta_{m,n} = \sum_n |c_n|^2$

$$\langle E \rangle = \sum_n |c_n|^2 E_n$$

verj., da za stanje Ψ izmerimo energijo E_n

$$\sum_n |c_n|^2 = 1 \quad \leftarrow \text{normalizacija}$$

⚠ pri vsaki meritvi izmerimo le eno vrednost E_n (pri veliko meritvah je $\langle E \rangle = \sum_n |c_n|^2 E_n$)

Zgled: popr. GK val. paketa

$$\Psi(x) = A e^{-x^2/4\sigma^2} e^{ik_0 x}$$



$$\frac{d\Psi}{dx} = A \left(-\frac{x}{2\sigma^2} + ik_0 \right) e^{-x^2/4\sigma^2} e^{ik_0 x}$$

$$\langle p \rangle = \int_{-\infty}^{\infty} \Psi^* \left(-i\hbar \frac{d}{dx} \right) \Psi dx = -i\hbar |A|^2 \int_{-\infty}^{\infty} e^{-x^2/4\sigma^2} \cancel{e^{-ik_0 x}} \left(ik_0 - \frac{x}{2\sigma^2} \right) e^{-x^2/4\sigma^2} \cancel{e^{ik_0 x}} dx =$$

$$= |A|^2 \hbar k_0 \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} dx - \underbrace{\int_{-\infty}^{\infty} x e^{-x^2/2\sigma^2} dx}_{=0} = \hbar k_0 |A|^2 \underbrace{\int_{-\infty}^{\infty} \dots}_{=1} = \hbar k_0$$

Kaj pa kin. en?

$$\langle T \rangle = \int_{-\infty}^{\infty} \Psi^* \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) \Psi(x) dx$$

$$\frac{d^2\Psi}{dx^2} = A \left(-\frac{1}{2\sigma^2} + \left(ik_0 - \frac{x}{2\sigma^2} \right)^2 \right) e^{-x^2/4\sigma^2} e^{ik_0 x} = A \left(-\frac{1}{2\sigma^2} - k_0^2 - \cancel{\frac{k_0 x}{\sigma^2}} + \frac{x^2}{4\sigma^4} \right) e^{-x^2/4\sigma^2} e^{ik_0 x}$$

bo iz
istil razl. kot prej

$$\langle T \rangle = -\frac{\hbar^2}{2m} |A|^2 \int_{-\infty}^{\infty} \left(-\frac{1}{2\sigma^2} - k_0^2 + \frac{x^2}{4\sigma^4} \right) e^{-x^2/2\sigma^2} dx =$$

$$= \frac{\hbar^2}{2m} |A|^2 \left[\left(\frac{1}{2\sigma^2} + k_0^2 \right) \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2} dx - \frac{1}{4\sigma^4} \int_{-\infty}^{\infty} x^2 e^{-x^2/2\sigma^2} dx \right] =$$

$$|A|^2 = \frac{1}{2\pi\sigma}$$

$$= \frac{\hbar^2}{2m} |A|^2 \left[\left(\frac{1}{2\sigma^2} + k_0^2 \right) \sqrt{2\pi} \sigma - \frac{1}{4\sigma^4} \sqrt{2\pi} \sigma^3 \right] = \frac{\hbar^2}{2m} \left(k_0^2 + \frac{1}{4\sigma^2} \right)$$

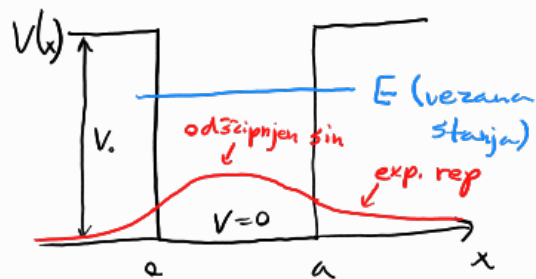
pričakovano: $\langle p^2 \rangle = \frac{\hbar^2 k_0^2}{2m}$

Heisenberg

→ $T \neq 0$ tudi če $k_0 = 0$ (ker smo delec lokalizirali)

Kvalitativno obnašanje VF vezanih stanj

• Ukrivljenost VF (končna pot. jama)



Zunaj jame: $-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V_0\psi = E\psi$

$$\frac{d^2\psi}{dx^2} = \underbrace{\frac{2m(V_0 - E)}{\hbar^2}}_{k^2} \psi$$

Rešitve:

$$\psi(x) = \begin{cases} e^{kx} & x < 0 \\ e^{-k(x-a)} & x > a \end{cases}$$

• -VF osn. stanja

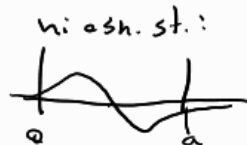
→ gladka funke.

Če $V(x) \neq \text{konst.}$

$$\underbrace{\frac{d^2\psi}{dx^2}}_{\propto p^2} \propto (V(x) - E)\psi$$

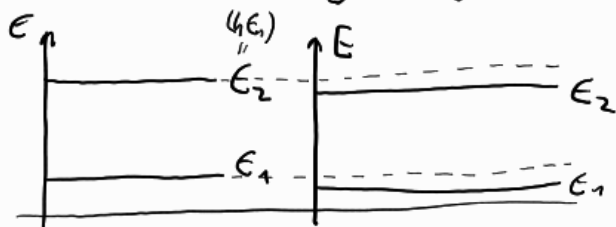
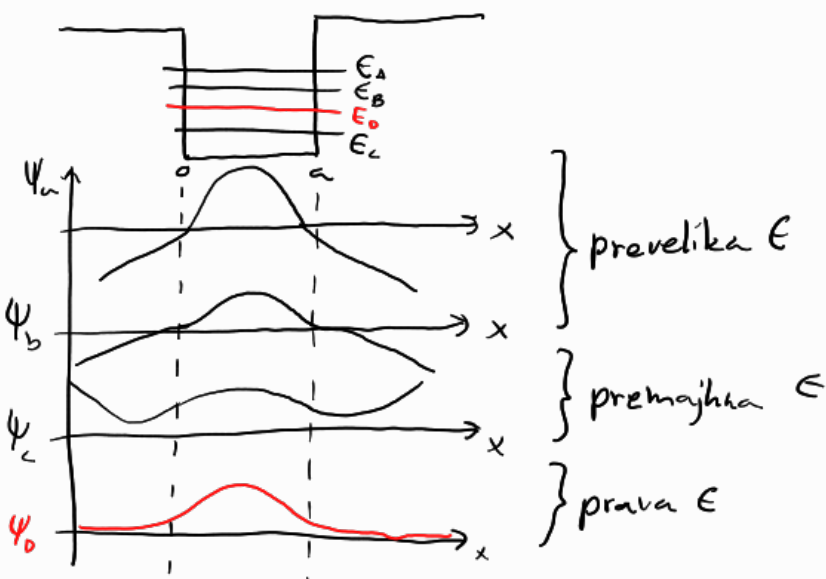
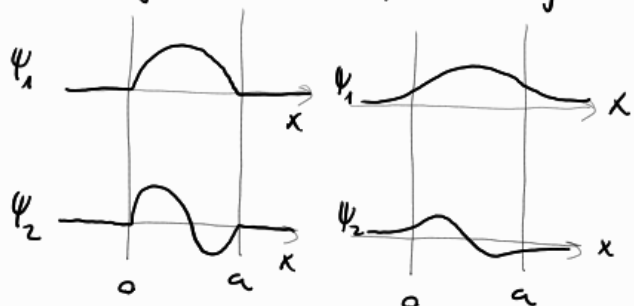
(ukrivljenost)

 → Osn. stanje ima VF, ki je najmanj ukrivljena →



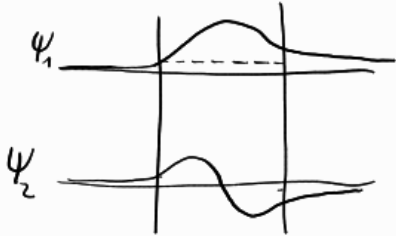
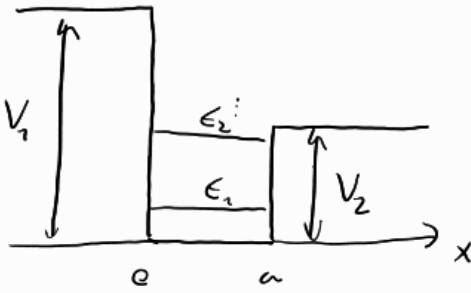
∞ jama

končna jama

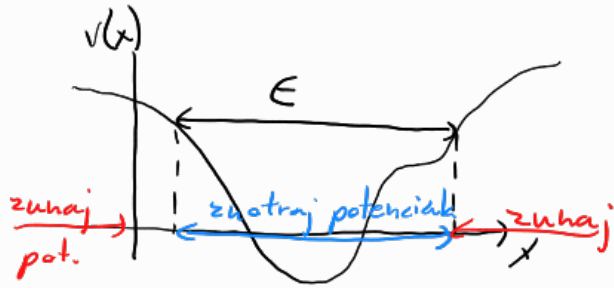


Zapletiti:

1)



2) poljubni $V(x)$



znotraj pot.: $\psi'' = -k^2 \psi$

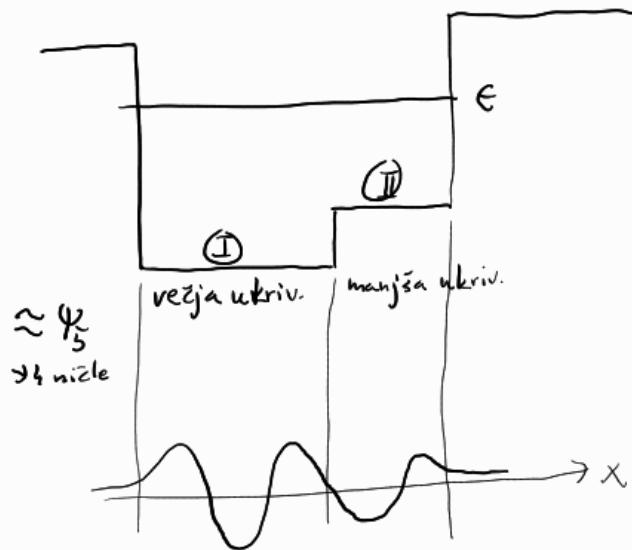
• oscilatorno, ukrivljenost proti osi x

zunaj pot.: $\psi'' = k^2 \psi$

• eksp. padanje, ukrivljenost stran od osi

Enačbe tipa $p\psi'' + q\psi' + r\psi = s$ so Sturm-Liouvilleovi problemi

→ N -to energ. stanje ima $N-1$ vozlov (ničel)



Amplitude znotraj:

$$\psi(x) = A \sin(kx + \phi)$$

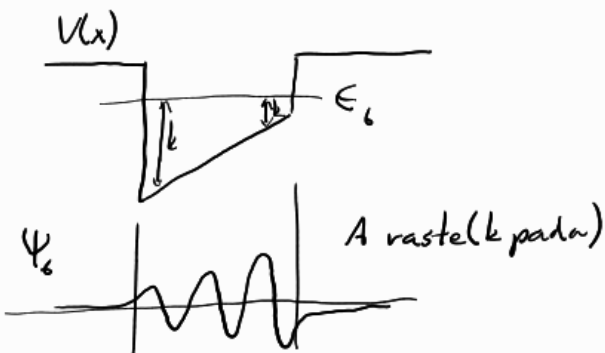
$$m = \frac{d\psi}{dx} = kA \cos(kx + \phi)$$

$12, +:$

$$A^2 (\sin^2 + \cos^2) = \psi^2 + \frac{m^2}{k^2}$$

$$A = \sqrt{\psi^2 + \frac{m^2}{k^2}}$$

← $k \uparrow$ (oz. $\lambda \downarrow$) $\Rightarrow A \downarrow$ in obratno



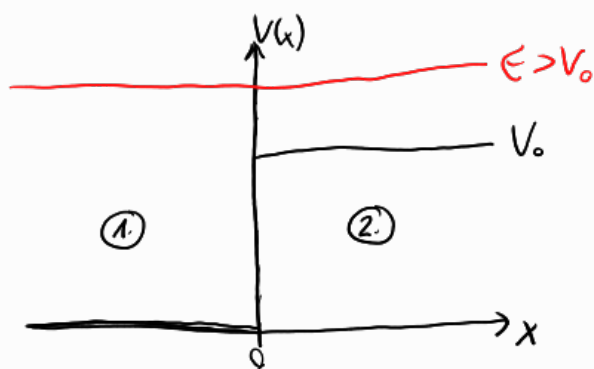
A raste (k pada)

($\exists$ točna rešitev: Airyjeve funkc.)

(za lin. pot. $V(x) \propto x$)

Nevezana stanja

Odboj na potencialni stopnici



$$\textcircled{1}: k_1 = \frac{\sqrt{2mE}}{\hbar}$$

$$\Psi_1 = Ae^{ik_1x} + Be^{-ik_1x}$$

$$\textcircled{2}: k_2 = \frac{\sqrt{2m(E-V_0)}}{\hbar}$$

$$\Psi_2 = Ce^{ik_2x} + \underbrace{De^{-ik_2x}}_{=0 \text{ (delec ne more priti iz desne)}}$$

$\Rightarrow$ (delec ne more priti iz desne)

gladko zlepimo:

$$\Psi_1(0) = \Psi_2(0) \quad \Psi_1'(0) = \Psi_2'(0)$$

$$A+B=C$$

$$ik_1A - ik_1B = ik_2C$$

2 enačbi, 3 neznanke?

- relevantni samo razmerji: $\left|\frac{B}{A}\right|^2$ in $\left|\frac{C}{A}\right|^2$

$$B = \frac{k_1 - k_2}{k_1 + k_2} A \quad C = \frac{2k_1}{k_1 + k_2} A$$

$$R = \text{odbojnost} = \frac{j_{\text{odbiti}}}{j_{\text{vpadni}}} = \left|\frac{B}{A}\right|^2 = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2$$

Spominimo se:

$$\Psi = Ae^{ikx}$$

$$j = \frac{\hbar}{2mi} \left(\Psi^* \frac{d\Psi}{dx} - \Psi \frac{d\Psi^*}{dx} \right) = |A|^2 \underbrace{\frac{\hbar k}{m}}_{=v} \propto |A|^2 k$$

(normiramo na tok j_0 , ker $\int_{-\infty}^{\infty} |\Psi|^2 dx = \infty$)

Prepustnost ($k_1 \rightarrow k_2$)

$$T = \frac{j_{\text{prep.}}}{j_{\text{vpadni}}} = \left|\frac{C}{A}\right|^2 \frac{k_2}{k_1} = \frac{4k_1k_2}{(k_1 + k_2)^2}$$

$$\text{velja } R+T=1$$

• Delci se odbijajo kljub temu, da $E > V_0$ (v nasprotju s KF)

• $R \propto (k_1 - k_2)^2 \Rightarrow$ pri skoku navzdol se odbije enako delcev kot pri skoku navzgor!



Obnašanje VF:



$E < V_0$:



Klas.: vsi delci se odbijejo

Kvantno:

$$\Psi_2 = C e^{ik_2 x}$$

$$\Psi_2 = C e^{-k_2 x}$$

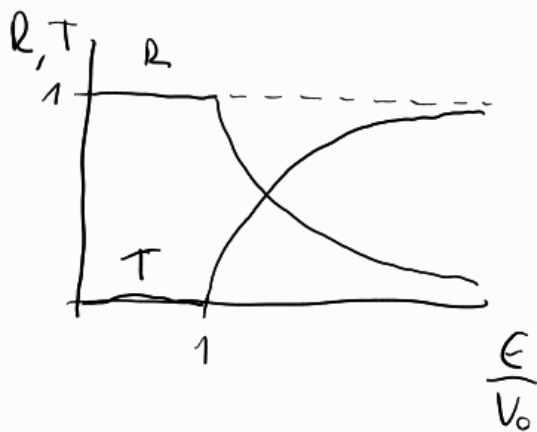
$$k_2 = \frac{\sqrt{2m(E-V_0)}}{\hbar} = i \frac{\sqrt{2m(V_0-E)}}{\hbar} \in \mathbb{C}$$

neniželna verj. gost. v prepovedanem območju

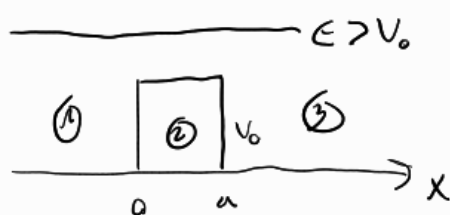
$$\text{ampak: } R = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2 = \left(\frac{k_1 - i k_2}{k_1 + i k_2} \right)^2 = 1$$

$\Rightarrow T=0$ prepustnost

- neniželna $|\Psi|^2$ v območju $x > 0$



Potencialna plast



$$\Psi = \begin{cases} \Psi_1(x) = A e^{ik_1 x} + B e^{-ik_1 x} & k_1 = \frac{\sqrt{2mE}}{\hbar} \\ \Psi_2(x) = C e^{ik_2 x} + D e^{-ik_2 x} & k_2 = \frac{\sqrt{2m(E-V_0)}}{\hbar} \\ \Psi_3(x) = F e^{ik_1 x} \end{cases}$$

gladko zvezano pri $x=0$, $x=a$:

$$\Psi_1(0) = \Psi_2(0)$$

$$\Psi_1'(0) = \Psi_2'(0)$$

$$A + B = C + D$$

$$ik_1 A - ik_1 B = ik_2 C - ik_2 D$$

$$\Psi_2(a) = \Psi_3(a)$$

$$\Psi_2'(a) = \Psi_3'(a)$$

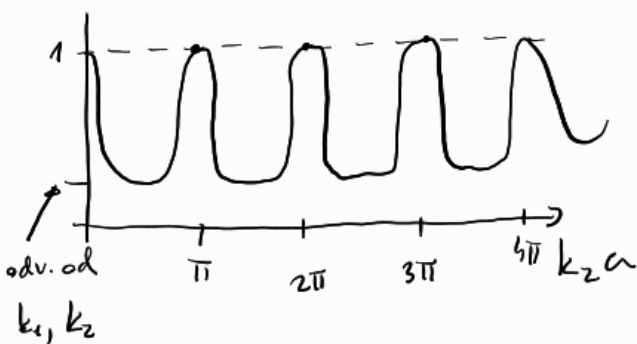
$$C e^{ik_2 a} + D e^{-ik_2 a} = F e^{ik_1 a}$$

$$ik_2 C e^{ik_2 a} - ik_2 D e^{-ik_2 a} = ik_1 F e^{ik_1 a}$$

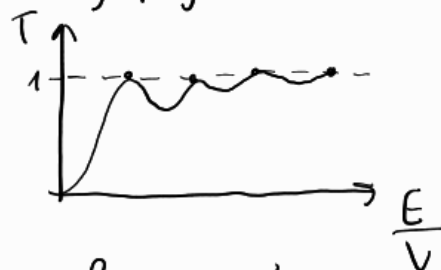
$$\begin{aligned} B &= \dots A \\ \Rightarrow C &= \dots A \\ D &= \dots A \\ F &= \dots A \end{aligned} \rightarrow T = \frac{j_{\text{prop.}}}{j_{\text{up.}}} = \left| \frac{F}{A} \right|^2 = \left(1 + \frac{1}{4} \left(\frac{k_2}{k_1} - \frac{k_1}{k_2} \right)^2 \sin^2 k_2 a \right)^{-1}$$

$$R = 1 - T = \left(\frac{C}{A} \right)^2 = \frac{\frac{1}{4} \left(\frac{k_2}{k_1} - \frac{k_1}{k_2} \right)^2 \sin^2 k_2 a}{1 + \frac{1}{4} \left(\frac{k_2}{k_1} - \frac{k_1}{k_2} \right)^2 \sin^2 k_2 a}$$

$$\text{če } k_2 a = n\pi \Rightarrow T = 1$$



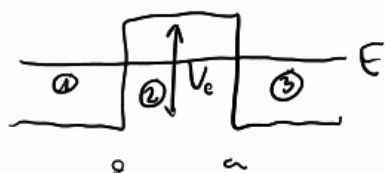
drugi pogled:



Ramsauerjev pojav
(najprej v žlahtnih plinih)

e^- s pravilno izbranimi E grejo nemoteno skozi oblak e^- v atomih vzorca

$$E < V_0$$



Ψ_1, Ψ_3 enako kot prej

$$\Psi_2 = C e^{ik_2 x} + D e^{-ik_2 x} \quad k_2 = iK; \quad K = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

$$\Psi_2 = C e^{-Kx} + D e^{Kx}$$

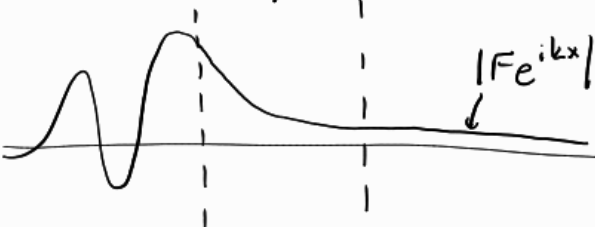
Spet kopino v $x=0, x=a$



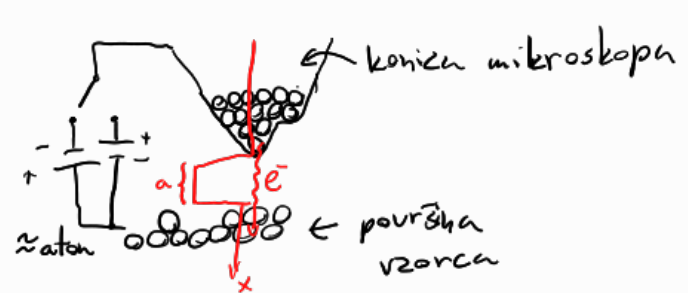
$$T = \left| \frac{F}{A} \right|^2 = \left[1 + \frac{\sinh Ka}{4 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right)} \right]^{-1}$$

V lim $Ka \gg 1$:

$$T \approx 16 \frac{E}{V_0} \left(1 - \frac{E}{V_0} \right) e^{-2Ka}$$



$|F e^{ikx}|^2 = |F|^2 = \text{konst.} \rightarrow$ osnova za vrstični e^- mikroskop



T (oz. tok prek reže) je eksponentno
odv. od a